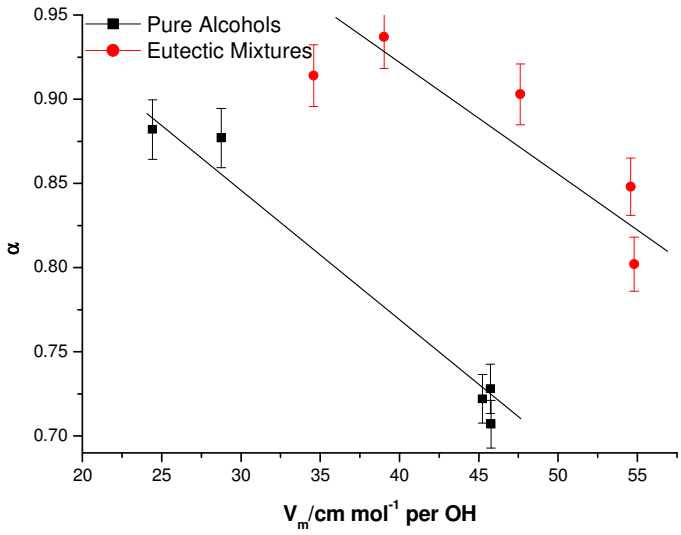


Figure 4.7: *Correlation between α and the molar volume per OH group*

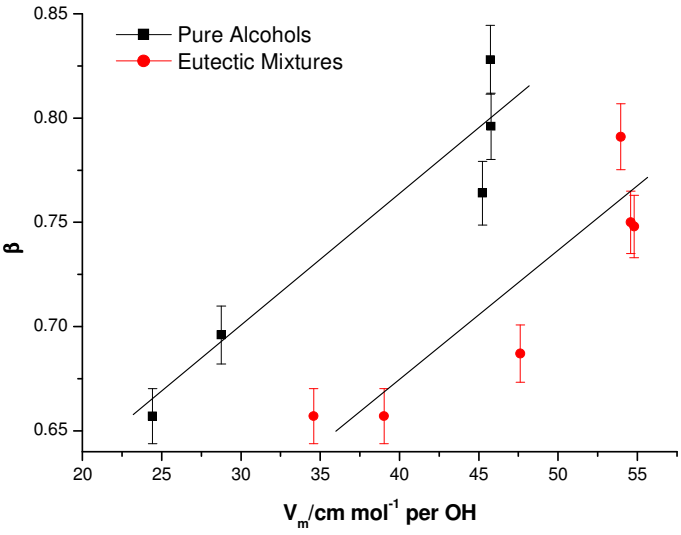


Figure 4.8: *Correlation between β and the molar volume per OH group*

However, there is negligibe correlation between π* and the molar volume per OH group showing that this simple model for the solvation in these eutectic systems is far from accurate.

4.2.3 Effect of Choline Chloride Concentration

To measure the effect of choline chloride concentration on the polarity of a system, the system most stable to containing concentrations of choline chloride above the eutectic point needed to be used. In this case glycerol based systems were used. Although it has been shown in Chapter 3 the eutectic point of ChCl and glycerol is in a 1:2 ratio, higher levels of ChCl can be added without phase separation at room temperature. **Figure 4.8** shows a roughly linear increase in $E_T(30)$ with ChCl concentration. Extrapolating this trend to 100% ChCl gives a value of approximately 59 kcal mol^{-1} which is comparable to that for ethylammonium nitrate (Table 4.4) which would seem reasonable.

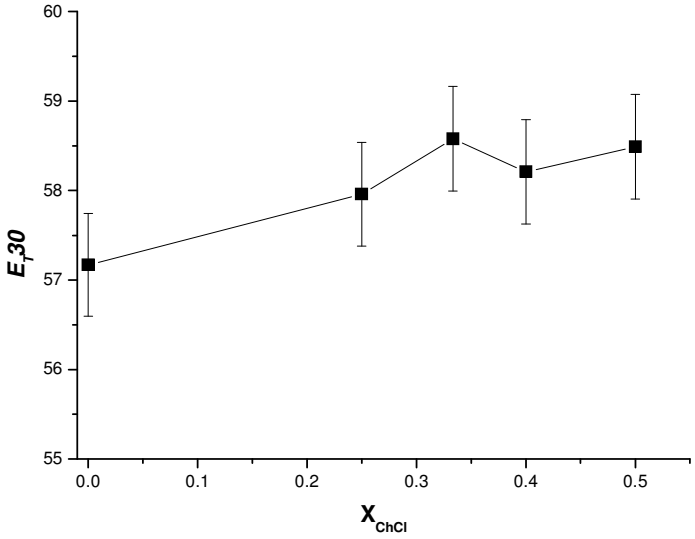


Figure 4.9: *$E_T(30)$ as a function of choline chloride concentration for the glycerol base systems*

Similar tends are observed for π*, α and β although the parameters are relatively insensitive to changes in polarity. The β parameter in particular does not appear to show significant changes with increasing chloride concentration. To some extent this is not surprising since the number glycerol OH groups is present in excess and will act as a better hydrogen bond donor than the indicator solute. It is proposed that in high polarity solvents solvatochromic indicators are insensitive to compositional changes because solvent-solvent interactions dominate over solvent-solute interactions.

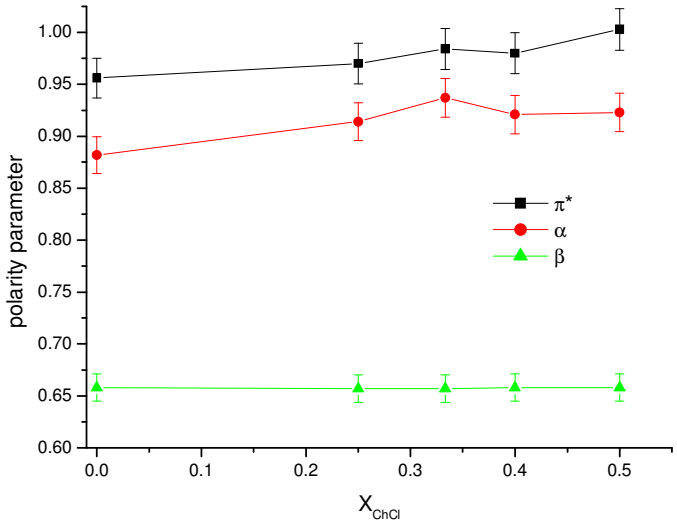


Figure 4.10: *Variation of π*, α and β with mole fraction of ChCl in glycerol*

Table 4.7: *Change in solvatochromic parameters as a function of ChCl concentration*

System	Δπ*	Δα	Δβ	Δ E_T (30)
ChCl:Glycerol 1:3	0.015	0.032	-0.001	0.79
ChCl:Glycerol 1:2	0.028	0.056	-0.001	1.41
ChCl:Glycerol 1:1.5	0.025	0.039	0.000	1.04
ChCl:Glycerol 1:1	0.048	0.041	0.000	1.32
ChCl:Ethylene glycol 1:2	0.080	0.026	-0.009	1.37
ChCl:1,4-Butanediol	0.259	-0.076	0.027	1.23
ChCl:1,3-Butanediol	0.121	0.074	-0.046	4.10
ChCl:1,2-Butanediol	0.095	0.140	-0.080	2.48

4.3 Conclusions

This work has shown that the interactions between the glycols and ChCl are complex. The polarity parameters of the mixtures are dominated by the properties of the glycols and there is little effect of ChCl on the hydrogen bond donor or acceptor properties of the glycerol based liquids. There is however a more drastic effect of ChCl

addition in the butanediol based systems, with the biggest change in polarity being observed in the 1,3-butanediol system. The addition of ChCl to glycerol is shown to increase the overall polarity of the system as more ChCl is added. The change in the polarisability parameter π* is roughly a linear increase, as can be seen in Figure 4.10, which would be expected since the number of ionic species increases. In the glycerol based systems the addition of ChCl shows negligible change in the hydrogen bond accepting parameter β. In the diol systems, the addition of ChCl causes a decrease in this parameter, which would be expected since the Cl^- will be interacting with the OH moieties, reducing the ability to further accept hydrogen bonds. There also appears to be some correlation between the density of OH bonds in the liquid and the $E_T(30)$, α and β polarity parameters showing that hydrogen bonding is important in controlling solvation in these systems.

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CHAPTER 5

Esterification of Glycerol in a Choline Chloride Based Deep Eutectic Solvent

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5.1 Introduction

5.1.1 Glycerol Esters

5.1.2 Project Aim

5.2 Development of Analytical Technique

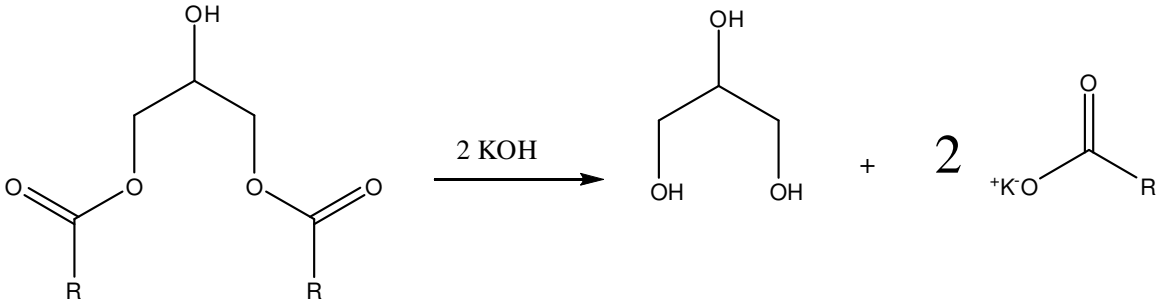
5.3 Results and Discussion

5.4 Conclusions

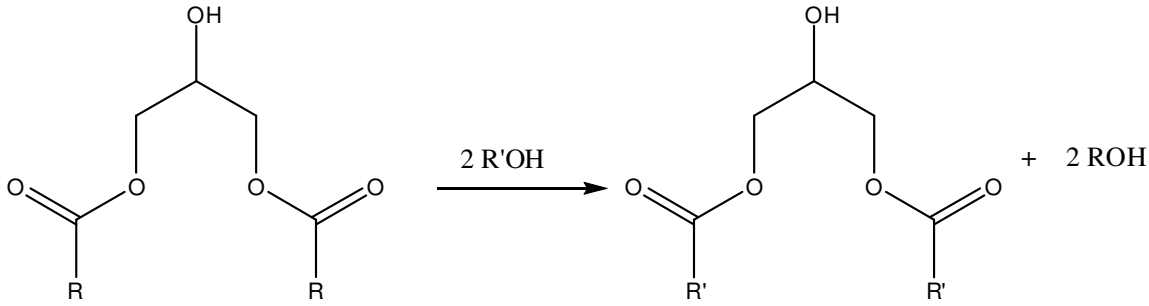
5.5 References

5.1 Introduction

Glycerol is a very important molecule as it is involved in a wide variety of biological processes in the production of fats, oils and phospholipids in both plant and animal cells. It is a compound that is produced on a large scale mostly as a waste material from the processing of oils. The saponification of oils to produce soap produces glycerol as does the transesterification of oil to produce biodiesel. Both of these processes are shown schematically in Figure 5.1



(1) Saponification of a diglyceride with KOH



(2) Transesterification of a diglyceride with R'OH

***Figure 5.1**: Saponification and transesterification schemes*

Current levels of glycerol production in Europe and the US are approximately 1 billion tonnes per year. The large worldwide scale of these processes has resulted in a huge surplus of glycerol which has very few uses. It is very difficult to burn and has little calorific value. It is used in some food and pharmaceutical applications, mostly as a thickening agent. There are many research groups and companies attempting to find applications for this material.

Glycerol can be used as a chemical feedstock from which to produce useful compounds including polyols, propylene glycol, acrolein, ethanol and epichlorhydrin.

5.1.1 Glycerol Esters

An alternative approach is to make mono- and di-esters of glycerol which can be used as emulsifiers, lubricants, surfactants, base materials for perfumes and antibacterial/antivirus drugs. Mono-esters are the most commonly used, although not exclusively, since they exhibit the highest surface activity of the acylglycerol types. More importantly, they offer very low toxicity to humans if ingested.

A considerable amount of work carried out on the esterification of glycerol.[1-10] Typically mono- and di-glycerides are formed by esterifying glycerol or by transesterifying the triglyceride, using acid or base catalysts respectively. Often these reactions are carried out in temperatures greater than 200 ºC. However, the selectivity for the manufacture of mono-ester or di-ester has been shown to be difficult in typical esterification reactions using acid catalysts or by base catlaysed transesterification of the triglyceride, where the product mixture contains the mono-, di- and tri-esters.[1-10]

There has been a lot of work carried out using zeolites as catalysts for the esterifaction of glycerol. The zeolites have been shown to offer catalytic selectivity for many organic reactions.[11-14] In work by Machado *et al.* Si/Al containing zeolites have been used to carried out the esterification of glycerol with lauric acid.[11] Machado *et al.* compared the reaction rates and selectivity using commercially available Beta, Y and

Mordenite zeolites. These zeolites contained different ratios of Si/Al and different pore sizes. Interestingly, it was observed that by choosing a zeolite with a high Si/Al, such as Beta, showed a higher selectivity could be obtained. This is attributed to two factors, firstly, the reduction in the Al sites results in stronger acid sites in the structure, and secondly, the hydrophobicity of the zeolite increase as the Si/Al ratio increases, which results in greater catalytic site activity. The smaller pore size in this system meant that di- and tri-ester formation is largely restricted. Al/Zr zeolites have alsovbeen employed for the formation of glycerol based esters under supercritical conditions.[12] Work from Van Rhijn *et al.* has shown that modifications of zeolites can increase the selectivity of the esterification products.[13-14] By adding sulfonic groups to the zeolites structure, an increase in the catalytic rate zeolite is observed, as well as an improvement on the selectivity.

Supercritical fluids have been employed as a reaction media in the esterifcation of glycerol.[12,15] Work by Moquin et al.[15] has used both supercritical CO_2 and N_2 in the esterification of oleic acid with glycerol. The use of supercritical CO_2 and N_2 is to act as a solvent in the reaction and to aid the recovery of the oleic acid from the product when the reaction was completed. It was reported that there was no difference in the rates or product distribution of the reaction between CO_2 and N_2.[15]

The esterifcation of glycerol with fatty acids has been investigated using ionic resins as a method of catalysis of the reaction.[16-17] These resins offer low operating temperatures and can act as shape selective material, due to their tunability to behave as gels or as macroporous resins, which improves the overall selectivity of the products. The catalytic properties of these sulfonic acid group containing cationic resins has been shown to be governed by the nature of the reagents. The adsorption of glycerol and fatty acid on the catalyst surface appears to show an effect on the catalytic activity. It is seen that several factors are involved in the activity of the reaction. The glycerol shows a show adsorption to the resin due to the strong interaction between OH moieties and the sulfonic groups on the resin, and this inhibits the rate of reaction of the esterification. As a result, excess glycerol further decreases the rate at which the reaction takes place hence increasing the selectivity towards the mono-ester. Work carried out by Rakotondrazafy *et al.*[17] has shown

that this method is not only suitable to cationic resins but anionic resins can be employed as well. The use of anionic resins gave rise to a high yield of monoglyceride.

Enzymes have been employed to form monoacylglycerols, and that the enzymes can be used to either esterify glycerol with the fatty acid or to selectively hydrolyse the triglyceride into the monoacylglycerol. Enzymes are known to exhibit regioselectivity and operate under very mild conditions for many organic synthetic reactions and so offer an alternative reaction pathway. Work by Monteiro *et al.*[18] has shown that by using a lipase enzyme derived from *Rhizomucor meihei* it is possible to preferentially form the mono-ester product. The reaction was carried out in a solvent of *n*-hexane/*t*-butanol (1:1 v/v) so the system was homogenous. Work by Bornscheuer *et al.*[19] has shown this using the lipase enzyme derived from *Pseudomonas cepacia*, both the esterification of glycerol with lauric acid and the transesterification of the trilaurin glyceride. The esterification reaction was carried out in a microemulsion system containing bis-(2-ethylhexyl)sulfosuccinate sodium salt (AOT)/isooctane, with an excess of glycerol: lauric acid (3:1). The mono-ester was formed preferentially in a ratio of ~2.3:1 mono- to di-ester. The isomeric form of these products was 1-monolaurin and 1,3-dilaurin. No evidence for esterification of the secondary OH moiety was observed. Lipase enzymes have also been shown to promote transesterification of triglycerides into biodiesel products such as propyl oleate.[20]

Solventless methods to esterify glycerol show litlle control over the selectivity of products formed. The work in this chapter shows the esterification of glycerol, containing various molar equivalents of choline chloride (ChCl), with lauric acid. This chapter investigates the addition of ChCl to glycerol, forming a Type III Deep Eutectic Solvent (DES). The idea behind adding ChCl to glycerol is to use the complexing relationship of the salt to form hydrogen bonds with the –OH moieties on glycerol to act partly as a *pseudo*-esterification site inhibitor and to alter the position of equilibrium of the reaction. For esterification to occur the OH moiety must be available to nucleophilic attack from the carbonyl group of the carboxylic acid. Other effects such as viscosity and concentration are also investigated.

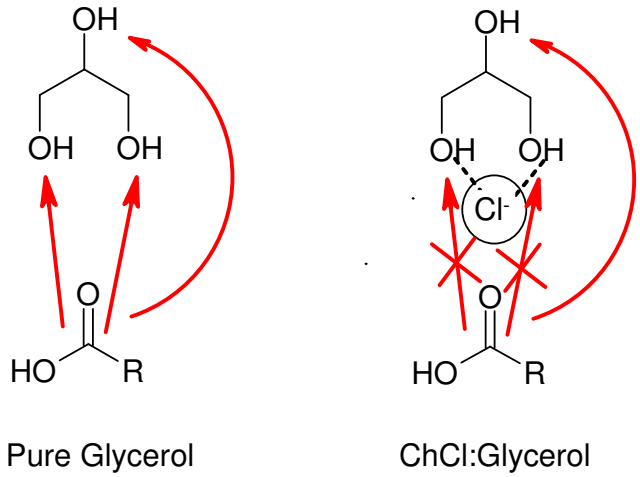


Figure 5.2: *Diagrammatic representation of possible site inhibition from ChCl addition*

The mechanism for the esterification of glycerol can be seen in Figure 5.3.

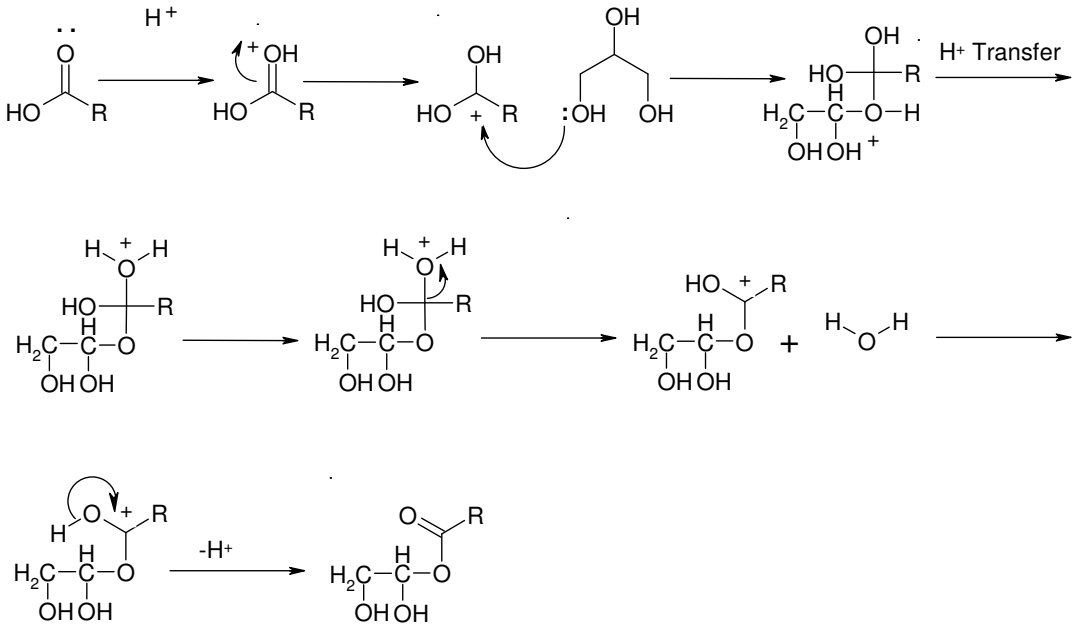


Figure 5.3*: Mechanism of Glycerol Esterification*

5.1.2 Project Aim

The aim of this section of the project was to use eutectics of glycerol with choline chloride to determine whether esterification reactions could lead to different product distributions and ideally selective production of mono- or di- esters of glycerol.

5.2 Development of Analytical Technique

Previous studies reported in the literature have used HPLC and GC to determine the product distribution from esterification reactions.[1-19] However using these protocols, poor resolution of the products was obtained and so a new methodology had to be developed. Gel permeation chromatography (GPC) has been used as an analytical method for this type of experiment before, but it is a much less common technique to be employed.[16] GPC is primarily used to separate large molecular weight products greater than 100,000 Daltons, such as polymers and proteins, however with specialist columns obtained from Polymer Laboratories, it has been possible to perform this method on molecules up to 2000 Daltons.

GPC is a chromatographic method that operates by performing size exclusion chromatography through a gel matrix. The main premise of this method is that larger molecules will not be trapped in the swollen matrix and therefore will elute faster from the stationary phase than the smaller molecules. In terms of the reaction being investigated, the tri-ester will elute first, followed by di-ester then mono-ester and finally the unreacted lauric acid. The technique however, is not designed to fractionate the various isomers of the mono- and di-esters as the column will only separate molecules of differing sizes. The eluent from the column is detected by refractive index. The solvent chosen to dissolve the products was THF. Sample spectra obtained can be seen in Figure 5.1.

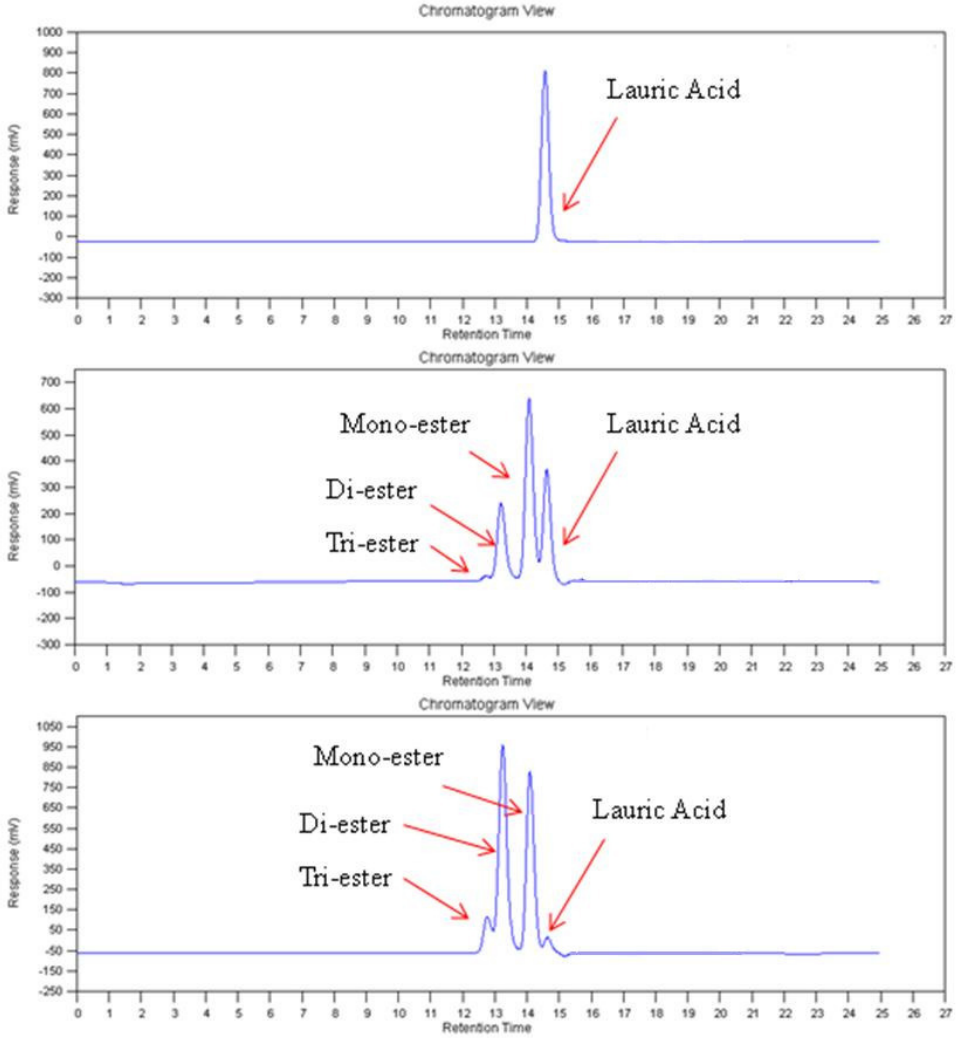


Figure 5.4: *Three GPC spectra, the first showing lauric acid, the second showing how the product distribution has altered over time and the third showing how reaction had further progressed.*

The data shown in Figure 5.4 is an example of the spectra obtained during analysis of the esterification reaction. In the first plot the data show the reaction at time zero, which only shows the presence of lauric acid. The second and third plots show how the spectra changes as the reaction progresses over time. It can be seen that the lauric acid peak decreases, while the mono-, di- and tri-ester peaks all increase.

To keep the peak heights inside of the range of the detector, several standards were run using lauric acid, as to determine the correct concentration of each sample needed, and it was determined that each sample vial would contain 0.02 g of product in 1.25 mL of THF. The temperature of the column was set at 40 ºC. The optimum flow rate was 1.0 mL/min. It should be noted that the ratio of peak areas does not alter by varying the concentration of the sample.

The estimated error in the data is believed to be of the order of +/-1% for all measurements so error bars have not been included in the following plots as these error bars are very close in size to the data markers.

5.3. Results and Discussion

The data in the following pages outlines the effect of initial acid concentration in mixtures containing various concentrations of ChCl. The aim of this work is to determine the effect of hydrogen bonding of Cl^- to the OH moieties on glycerol. It is envisaged that the strength of the H-bonding will affect the reactivity of glycerol and this could change the product distribution and/or the kinetics of the reaction.

The esterification of the triol can be broken down into three elementary steps each of which is an equilibrium. Analysis of such a complex process is clearly complicated and therefore only a simple quantitative analysis of the first step is carried out. Attempts to fit all of the data were unsuccessful due to the inaccuracy associated with the analytical technique.

The scheme for this reaction is as follows:

[glycerol] + [lauric acid] ⇌ [monoester] + [water] $$K_1 = \frac{[monoester][water]}{[glycerol][lauric\ acid]} \quad \textbf{(5.1)}$$

[monoester] + [lauric acid] ⇌ [diester] + [water] $$K_2 = \frac{[diester][water]}{[monoester][lauric\ acid]} \quad \textbf{(5.2)}$$

[diester] + [lauric acid] ⇌ [triester] + [water] $$K_g = \frac{[triester][water]}{[diester][lauric\ acid]} \quad \textbf{(5.3)}$$

The plots shown in Figure 5.5 show the progression of the esterification of lauric acid in various ChCl: glycerol DES mixtures and also pure glycerol. In each system, 60 cm^3, (0.82 mol) of glycerol was used and corresponding molar amount of ChCl needed to make up the desired ratio was added. It should be noted that the addition of ChCl changes the volume of the system and so the data have not been analysed in terms of the concentration, but rather the number of moles of each reagent. In each case the acid catalyst added was trifluoroacetic acid, 6 cm^3, which was chosen as it is a monoprotic acid that has been shown previously to form a DES with ChCl and does not appear to react with ChCl. In these systems 0.1 moles of lauric acid was added to systems. The reactions were all carried out at 150 °C, in a sealed vessel. As lauric acid and the respective esters are insoluble in glycerol and the glycerol based DES's, aliquots were taken of the top organic layer, which at this temperature was liquid. The isomeric form of the mono- and di-esters were not investigated in the analysis. Only the formation of the mono- and di-esters were investigated. The ChCl: glycerol 1:3 data set is not available for the 0.1 moles data.

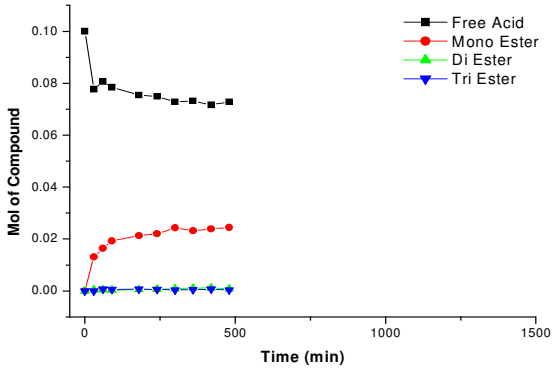
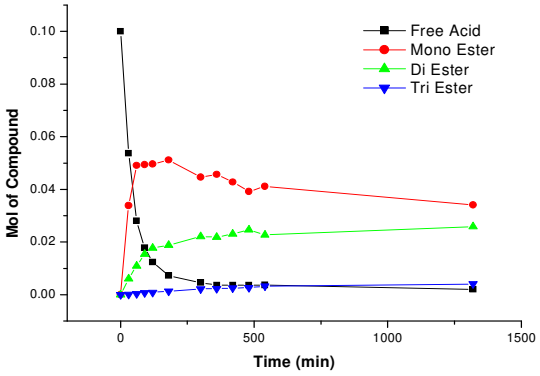
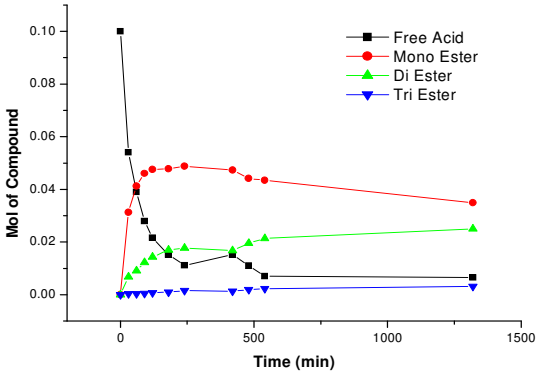
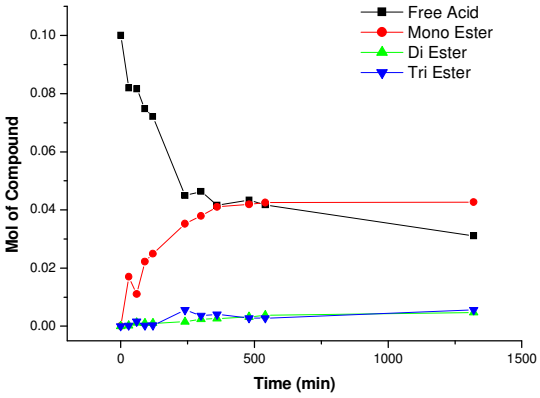
<table>
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</td><td>No reliable data obtained</td></tr>
<tr><td>Glycerol</td><td>ChCl: glycerol 1:3</td></tr>
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<tr><td>ChCl: glycerol 1:2</td><td>ChCl: glycerol 1:1.5</td></tr>
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<tr><td>ChCl: glycerol 1:1</td><td></td></tr>
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Figure 5.5: *Esterification Data of 0.1 moles of Lauric Acid in Various ChCl: Glycerol DES Mixtures*

The data in Figure 5.5 show the addition of ChCl does have a marked effect on the product distribution of the reaction. These systems are operating in large excess of glycerol, with a greater than 8:1 ratio of glycerol: lauric acid. The mono-ester would be the most abundant product predicted due to the large excess of glycerol. This is indeed the case with pure glycerol where negligible amounts of di- and tri-esters formed. It also shows that after 8 hours there is still approximately 70% of the free acid remaining. This is a surprising result since there is such a large amount of glycerol available for the lauric acid to react with but this could be due to the high viscosity of the system.

The 1:1 ChCl: glycerol mixture shows a faster consumption of the lauric acid than the pure glycerol system, resulting in a much greater formation of the mono-ester. It should be noted that twice as many moles of products are made when ChCl is added which clearly negates any effect of dilution. The 1:1.5 and 1:2 systems show similar overall conversion to the monoester but noticeably the monoester concentration goes through a maximum due to its use in reaction (5.2). The concentration of the diester is significantly increased with respect to both the pure glycerol and the 1:1 ChCl systems. The rate of consumption of the lauric acid follows 1:2 ≈ 1:1.5 > 1:1 > glycerol. This could be due to the *pseudo* site inhibition, but a more logical explanation is that the process is mass transport controlled.

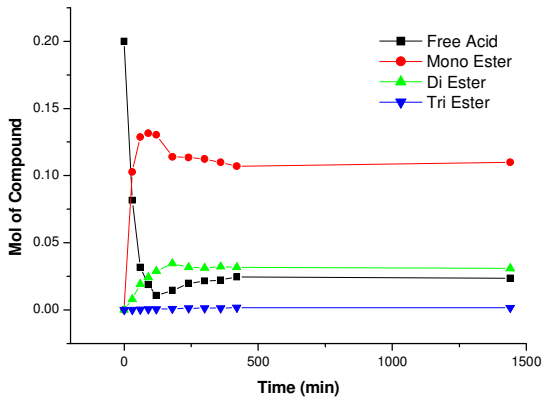
This is an oversimplification because the system is in effect an emulsion, however because the catalyst, glycerol and ChCl are in one phase while the lauric acid is in the other it is still important to consider the mass transport of species to and from the interface. Another issue that cannot be overlooked is the activity of the catalyst. In aqueous solutions pH will be dependent upon ionic strength. This should also be the case in an ionic liquid although it is an area that has not been significantly studies.[21]

The results shown in Figure 5.5 indicate that the addition of choline chloride alters the product distribution of the esterifcation of glycerol and lauric acid. The 1:2 mixture shows that it is possible to form the di-ester product in relatively large amounts considering that initial ratio of glycerol: lauric acid is very high. Deviating slightly to a ratio of 1:1.5 barely alters this product distribution, however increasing to a ratio of 1:1 the product

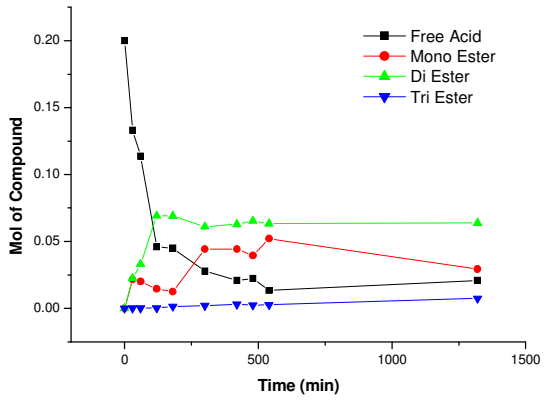
distribution more closely resembles that of the of the pure glycerol system. There are likely to be many factors governing how these reactions progress. On one hand the addition of ChCl to glycerol results in a lower viscosity, which would increase the mass transport in the system.

This section has shown that at low acid concentrations and high ChCl concentrations the monoester can be selectively produced at a significantly higher rate than in pure glycerol.

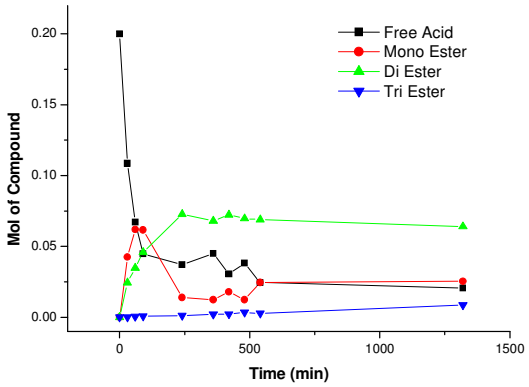
An additional point worth discussing at this juncture is that the choline cation contains an OH moiety which in principle could be esterified. Despite numerous experiments no such esters were observed. Bell *et al.* also studied esterification reactions in choline based ionic liquids and deep eutectic solvents. The choline cation was found to be very unreactive and it was only when strong Lewis acids such as $ZnCl_2$ were present with acetic anhydride that any significant acylation took place.[22]



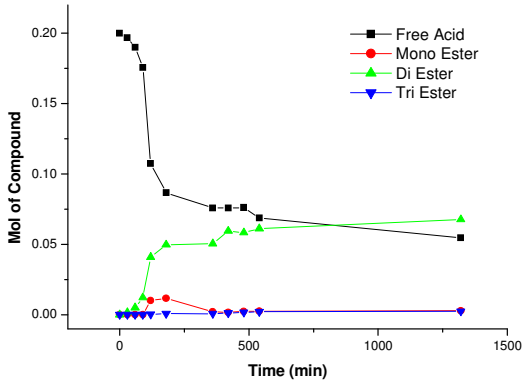
Glycerol



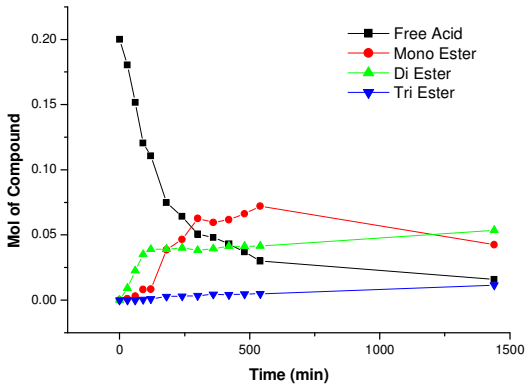
ChCl: Glycerol 1:3



ChCl: Glycerol 1:2



ChCl: Glycerol 1:1.5



ChCl: Glycerol 1:1

Figure 5.6: *Esterification Data of 0.2 M Lauric Acid in Various ChCl: Glycerol DES Mixtures*

The data shown in Figure 5.6 is directly comparable to that in Figure 5.5, but with twice the concentration of lauric acid added. The first point to note is that in the pure glycerol system the rate at which lauric acid is used up is considerably faster than in the previous case. It is also noteworthy that the free acid concentration reaches a minimum before reaching a constant value of c.a. 0.02 moles. This is characteristic of system at equilibrium *i.e.* the back reaction of equation (5.1) becomes significant. It can also be seen that almost no tri-ester forms. The addition of 0.33 eq. of ChCl causes a decrease in the rate of acid usage, a decrease in the yield of mono-ester but a significant increase in the concentration of di-ester and tri-ester. The same trends are observed for the 1:2 and 1:1.5 mole ratio mixtures. Most importantly in the 1:1.5 mixture the di-ester was effectively the only product produced. This shows that the aim of this section of the work is possible, namely that selective alkylation can be achieved. The isomerisation of the product was not one of the issues of the project although other groups have elucidated this using nmr spectroscopy.[1-10] It is also evident from the data in Figure 5.6 that the ChCl systems have not reached equilibrium within the timescale of the experiment. The reason appears to be that the tri-ester is still being formed, albeit at a slow rate.

In the equimolar ChCl: glycerol mixture the rate of conversion decreases and the monoester is the favoured product. The same trend was observed using 0.1 moles of free acid. This hints at the possibilty that the *pseudo* site inhibition may occur at high ChCl concentrations.

Overall the results show a different set of trends than the data seen in Figure 5.5 *i.e.* the trends can not simply be explained by mass transport. As would be expected more di-ester products are formed as a direct result of more lauric acid being available. In this experiment, the pure glycerol system shows a much faster consumption of lauric acid than any of the ChCl based systems which is in direct contrast to the trend shown in Figure 5.5. It should be noted that while the reaction reaches equilibrium rapidly the product distribution is not selective for the mono-ester as seen above.

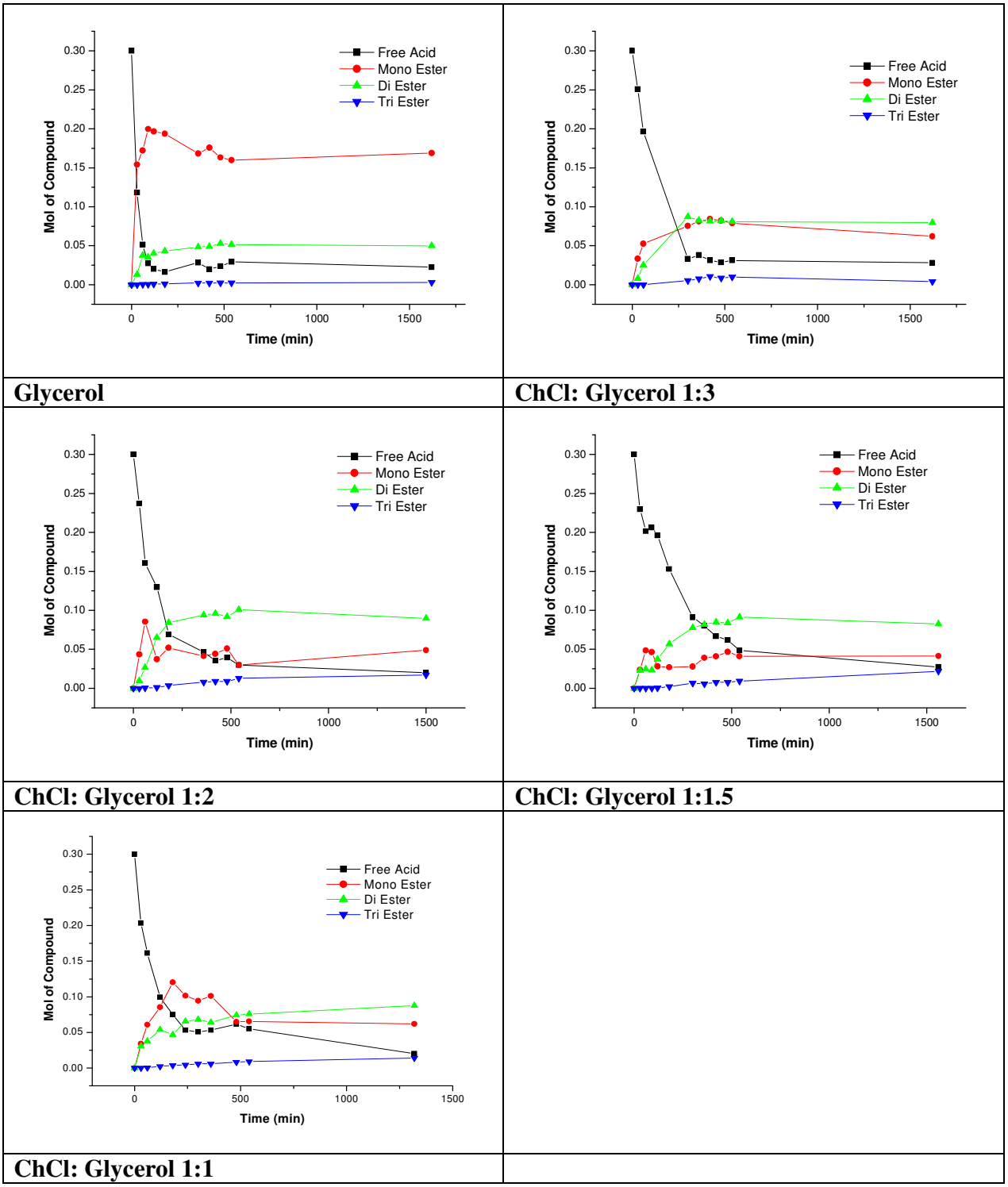


Figure 5.7: *Esterification Data of 0.3 M Lauric Acid in Various ChCl: Glycerol DES Mixtures*

The data in Figure 5.7 are all carried out using 0.3 M lauric acid. The data for the pure glycerol system shows similar trends to that seen in Figure 5.6 with the trend in the product distribution mono- > di- > tri-ester. The product distribution also appears to reach a constant value rapidly. If this is a true equilibrium then the equilibrium constants should be independent of acid concentration. Table 5.1 lists the equilibrium constants, K_1, K_2 and K_3 as a function of concentration.

Table 5.1.: *Equilibrium constants for the esterification reaction in pure glycerol*

Lauric acid/ moles	**K_1**	**K_2**	**K_3**
0.2	1.58	0.41	0.02
0.3	2.43	0.79	0.04
0.4	2.24	0.62	0.04
0.6	2.14	1.11	0.15

Table 5.1 shows that with the slight exception of 0.2 molar data, the values for K_1 are very close and suggest that in each reaction the system has reached equilibrium. The values for K_2 are less accurate but do hint at having reached equilibrium, as do the K_3 data. While these reactions qualitatively appear to be equilibria, it is difficult to gain any quantitative information from the data.

The 1:1 system shows a faster rate of formation of the mono-ester than in the 0.2 M system, as would be expected since the amount of lauric acid is greater. In all of the ChCl containing systems the amount of di-ester formed is approximately constant (0.01 moles). This is significantly higher that the pure glycerol system. All of the systems containing ChCl do not appear to have reached equilibrium, again because the tri-ester appears to form slowly and removes di-ester from the system.

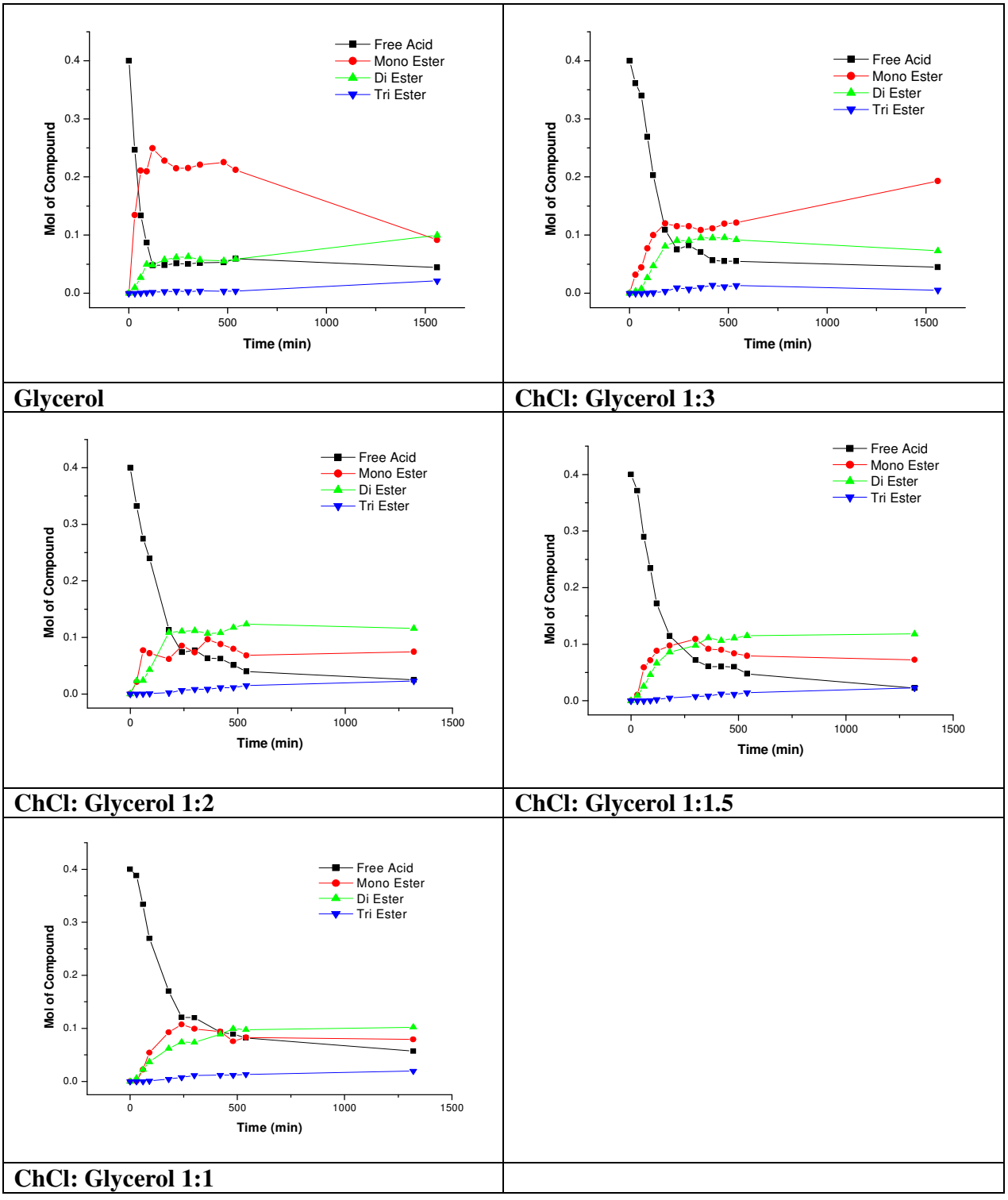
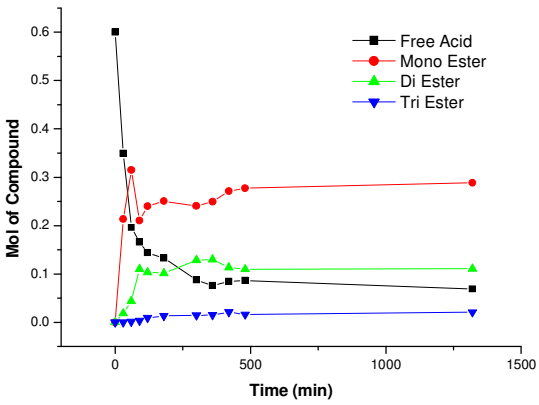


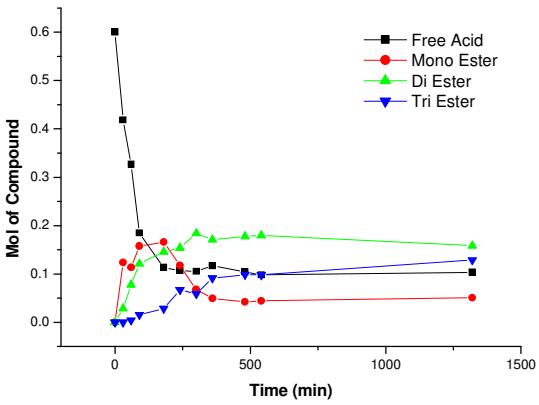
Figure 5.8: *Esterification Data of 0.4 M Lauric Acid in Various ChCl: Glycerol DES Mixtures*

Increasing the lauric acid concentration to 0.4 moles results in a similar product distribution to that found in the 0.3 moles and as Table 5.1 shows the equilibrium constants for pure glycerol remain relatively similar. As the lauric acid concentration increases the data for the ChCl containing systems approach equilibrium. The equilibrium constants are shown in Table 5.1 In all of the pure glycerol systems the equilibrium constants are in the order $K_1 > K_2 > K_3$ whereas for the ChCl containing mixtures the order is $K_2 > K_1 > K_3$. The only way in which the equilibrium can be driven to the right is if one of the components is removed or more accurately if it decreases its activity in the system. One possible explanation for the data observed in Tables 5.1 and 5.2 is that the presence of ChCl causes a decrease in the activity of water due to hydrogen bonding between the chloride anion and the water molecules.

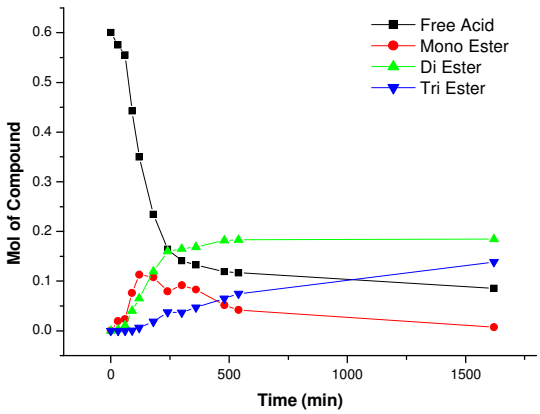
One complication is that the reaction is carried out at 150 ^{o}C which would normally be enough to drive the water from the reaction vessel however in these experiments the reaction vessel was closed and so the ChCl would only affect the concentration of water vapour in the headspace. Nevertheless, inspection of the data in Table 5.2 suggests that both K_2 and K_3 increase as the ChCl concentration increases which supports this idea. Furthermore the central premise of this work was that ChCl could act as a pseudo-site inhibitor. This could either prevent esterification, which is clearly not the case, or it could reduce the activity of glycerol in equation (5.1). If this were the case, then di- and tri-esterification would be preferred, which is indeed what appears to occur.



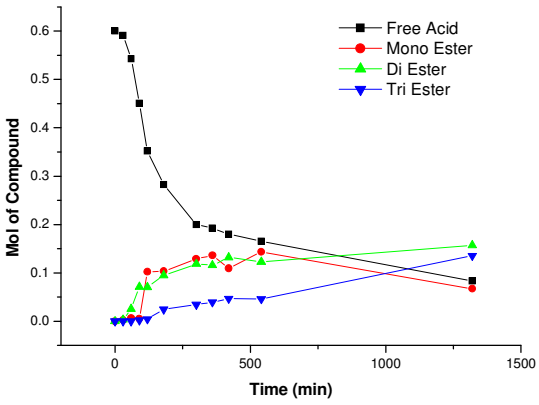
Glycerol



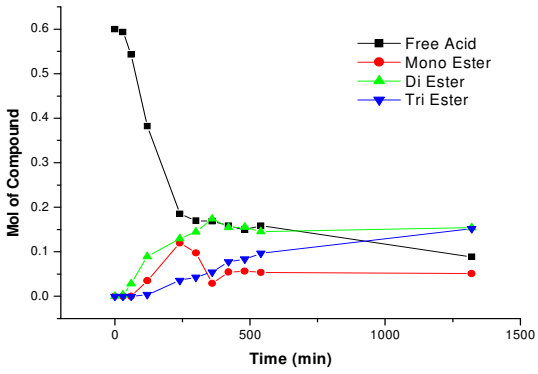
ChCl: Glycerol 1:3



ChCl: Glycerol 1:2



ChCl: Glycerol 1:1.5



ChCl: Glycerol 1:1

Figure 5.9: *Esterification Data of 0.6 M Lauric Acid in Various ChCl: Glycerol DES Mixtures*

The results for the system containing 0.6 moles of ChCl superficially support all of the discussions above. The concentration of the mono-ester is up to ten times larger than that observed in some of the ChCl containing systems. Given that glycerol is still present in a large excess it is important to notice that the overall lauric acid concentration is approximately the same at the end of each experiment. While this could be viewed in terms of decreased glycerol activity in the ChCl an alternative view could be that the ChCl breaks up the structure of the glycerol and therefore actually enables further reaction on the OH moieties that could normally be inter-molecularly hydrogen bonded. It is probably a mixture of both effects that lead to the observed product distribution.

***Table 5.2.**: Equilibrium constants for the formation of the mono-, di- and tri- esters for all ChCl based systems shown in Figures 5.5 -5.9*

Ratio of ChCl:Glycerol	Lauric acid/ moles	K_1	K_2	K_3
1:3	**0.2**	1.038	1.26	0.053
	0.3	1.053	1.080	0.13
	0.4	1.28	0.97	0.14
	0.6	0.64	2.58	1.41
1:2	**0.1**	1.43	0.79	0.11
	0.2	0.24	0.66	0.026
	0.3	0.40	1.36	0.174
	0.4	1.01	1.83	0.22
	0.6	0.44	1.91	0.78
1:1.5	**0.1**	0.48	0.21	0.021
	0.2	0.006	0.16	0.005
	0.3	0.31	0.70	0.070
	0.4	0.97	1.40	0.17
	0.6	0.90	0.77	0.290
1:1	**0.1**	0.077	0.007	0.005
	0.2	0.58	0.34	0.040
	0.3	0.44	0.50	0.061
	0.4	0.52	0.61	0.082
	0.6	0.41	1.11	0.74

Equilibrium constants have been calculated for the ChCl based systems, and are shown in Table 5.2.

Attempts were made to analyse the initial rate data using both integrated rate equations and the initial rate method, however the scatter of both the pseudo-first order analysis and the log rate vs. initial concentration plots were such that the results were statistically meaningless.

5.4 Conclusions

In this Chapter the solventless esterification of glycerol with the addition of ChCl has been studied for the first time. This work has shown that the addition of ChCl to glycerol can significantly affect the product distribution for the esterification of glycerol with lauric acid. Specific mono-esters and di-esters can be achieved by varying the liquid components. This site specific esterification is thought to occur due to the ability of ChCl to break up the hydrogen bonding in glycerol, but also to interact with water molecules produced in the reaction. The ChCl also decreases the viscosity of the liquid and at low acid concentrations when the system has not reached equilibrium, it is suggested that the process is mass transport controlled.

5.5 References

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